# North vs. South Campus Bruinwalk Reviews

By Jeremy, Muzhen, Suraj, Meiyi, Laura, & Ena



# Researchers



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# Vision



## Questions

1. Is there a significant difference in predicting the overall rating of a class between North and South campus classes?
2. What are critical factors in forecasting the rating?



## Purpose

To understand potential disparities in student satisfaction, instructional quality, and allowing targeted improvements for UCLA faculty to enhance educational experiences.

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What can we conclude & what would we do next?

# Abstract

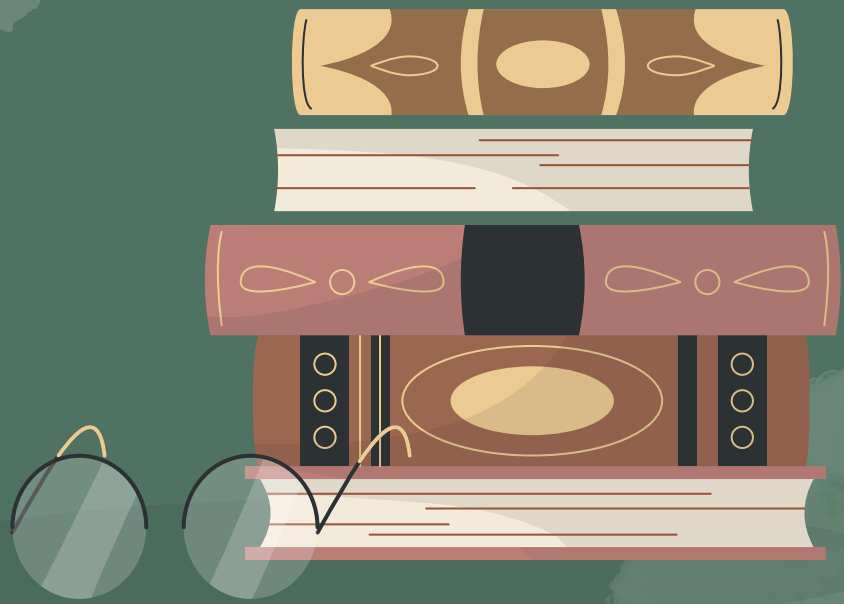
In this study, we explore the profound impact of sentiment analysis as a crucial tool to observe students' academic experience and achievements, particularly focusing on their perceptions of faculty across UCLA. We utilized student reviews and sentiments sourced from BruinWalk—a comprehensive repository of professor assessments and course evaluations, as a medium to investigate the predictive power of various machine learning models regarding course ratings. We are interested in comparing STEM-related fields, including Chemistry, Mathematics, Physics, and Statistics, to non-STEM fields including Communication, Economics, Political Science, and Psychology. In doing so, we aim to work towards enhancing the quality of education and fostering a conducive learning environment within higher education



# 01

# Sourcing the Data

Scraping Bruinwalk



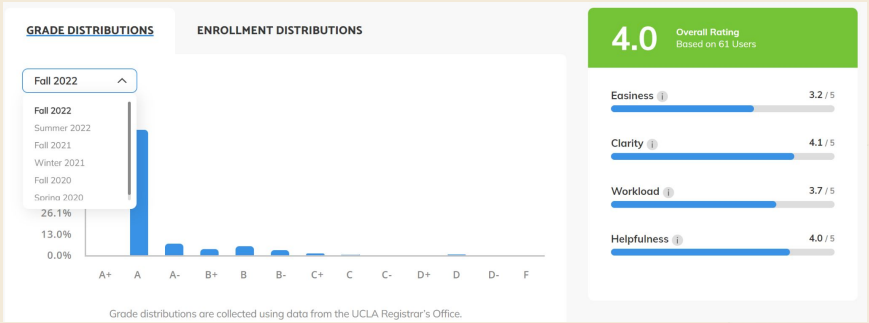
# Methodology

Packages used:

- bs4
  - BeautifulSoup
    - Create html file of website to extract text
    - Use text parsing techniques to isolate desired parts of webpage
- selenium
  - webdriver
    - Used to simulate mouse clicks

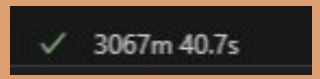
```
class_page = requests.get(URL)
soup = BeautifulSoup(class_page.content, "html.parser")
```

Iterate through: department(s)
→ each class
→ each professor that taught the class
→ every year the class was taught
→ every page of comments

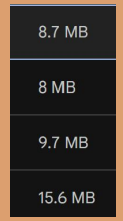


# Challenges

- Inexperience
- Run times
  - Projected to take weeks to iterate through every possible combination
  - Limited ability to optimize since website load time is one limiting factor
- Complicated interactions
  - Have to iterate through many instances
- Relatively large file sizes
  - Inefficient code
  - A lot more classes than expected




```
driver.get(URL)
quarter_box = driver.find_element(By.ID
actions = ActionChains(driver)
actions.key_down(Keys.CONTROL).click(qu
#search_quote = "//li[contains(text(),"
search_quote = f"//li[contains(text(),
time.sleep(1)
#print(search_quote)
option = driver.find_element(By.XPATH,
#time.sleep(2)
option.click()
time.sleep(0)
page_source = driver.page_source
repetition = i
soup = BeautifulSoup(page_source, "html
```



# 02 Processing

Cleaning and Basic Exploration



# Data Cleaning

**Phase 1** → Consolidated 8 department-specific datasets into one, converted all rating and grade percentage values to numeric, and replaced textual “N/A” with NA for proper missing value handling.

**Phase 2** → Added column “Dept.Type” to classify courses as STEM or non-STEM and “Dept” for department categorization based on class codes.

**Phase 3** → Eliminated rows completely missing relevant data and created a separate dataset for sentiment analysis with unique reviews.

**Phase 4** → Removed rows with absent grade data or duplicate class and grade entries.

**Phase 5** → Calculated and integrated weighted grade distribution averages back into GPA dataset to generate an “Average_GPA” column.

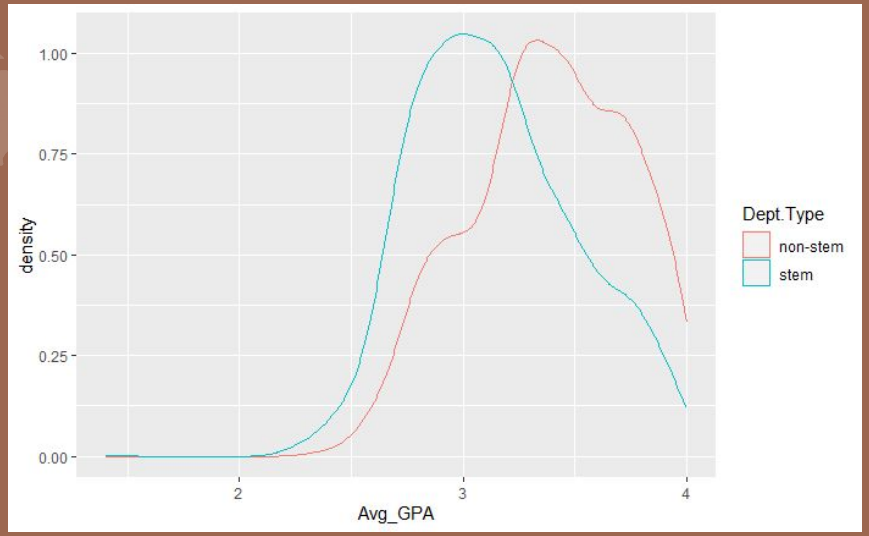
**Phase 6** → Refined dataset from ~157k entries to ~2.5k entries, with a final non-STEM class count of 939 and STEM count of 1,535.

# Variables Considered

- **Overall Rating:** The overall rating of the course from 0 to 5
- **Easiness:** The easiness rating of the course from 0 to 5
- **Clarity:** The clarity rating of the course from 0 to 5
- **Workload:** The workload rating of the course from 0 to 5
- **Helpfulness:** The helpfulness rating of the course from 0 to 5
- **Average GPA:** The weighted average GPA from 0 to 4 of a course in a given quarter. It is calculated using the percentage of people who received a grade multiplied by the UCLA registrar numeric GPA associated with a letter grade (4.0 for A and A+, 3.7 for A-, 3.3 for B+, etc.)
- **Department Type:** Categorical variable of departments indicated as STEM or non-STEM

# Data Exploration

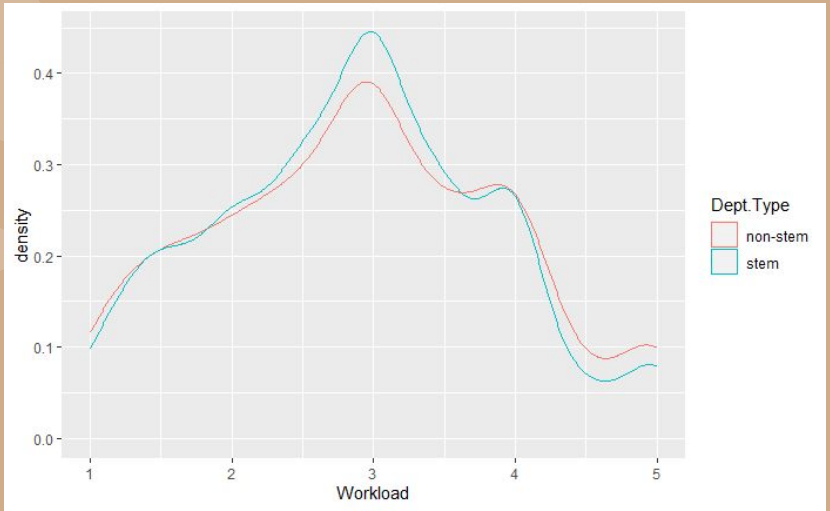
Average GPA density plots




- Separated by department type
- Most visually significant difference is in average GPA
- Both STEM and non-STEM have similar variable distributions

# Data Exploration

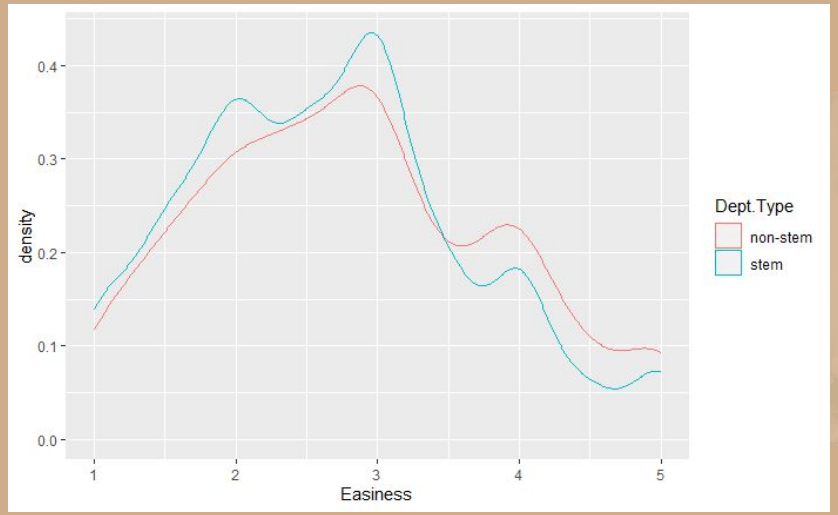
Workload rating density plot separated by department type




- Both models generally have a similar density plot with slight differentiation.
- Non-STEM departments have a slightly lower peak density of around 0.38 compared to STEM departments with a peak density of approximately 0.44
- There are several points of overlap around 1.5, 3.6, and 3.9. However, there is no large distinction between both department types.

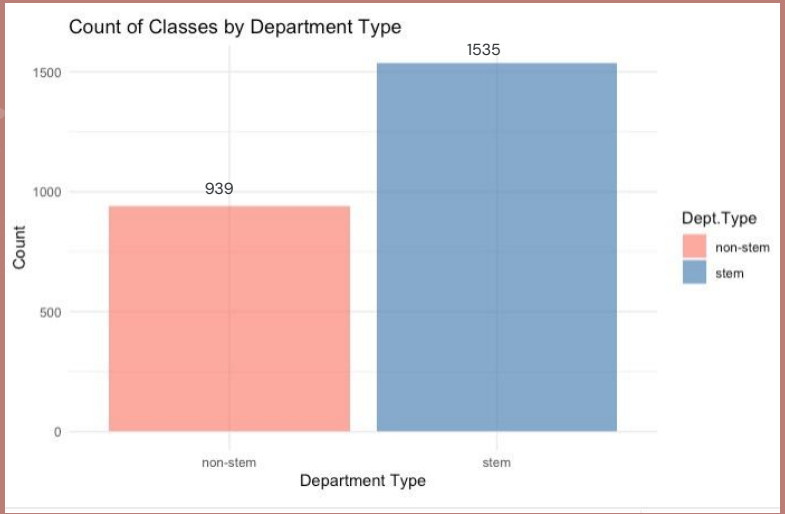
# Data Exploration

- From ranks 1 to 3, it appears that STEM departments have a generally higher density. However, beyond rank 3, non-STEM departments begin to have a higher density than STEM.
- Non-STEM departments have a lower peak density of around .38 compared to STEM departments with a peak density of around 4.3.




Easiness rating density plot separated by department type

# Data Exploration



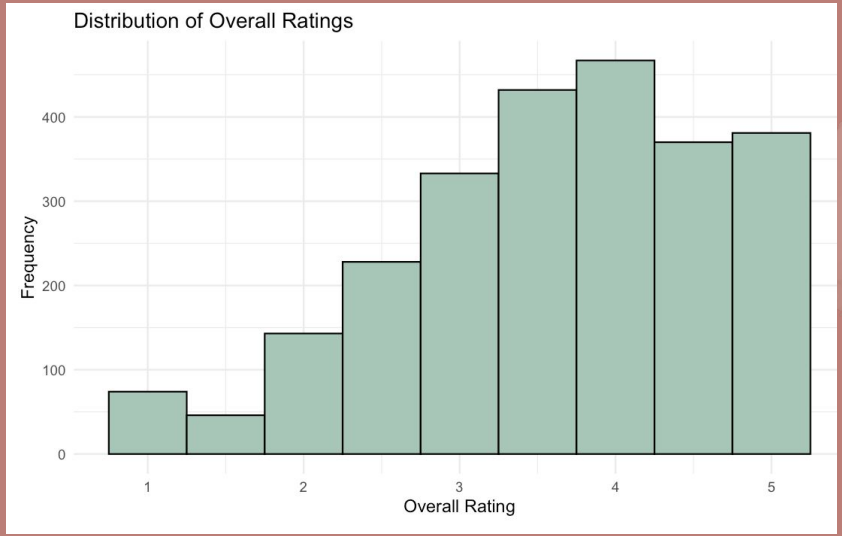

Count by Department Type

- Count of non-STEM classes (Communication, Political Science, Psychology, Economics) is lower than STEM classes (Chemistry/Biochemistry, Math, Physics, Statistics)

# Data Exploration

Distribution of Overall Ratings

- Distribution of overall rating has majority between 4 and 5.
- Overall shape is skewed left given peaking frequency of higher ratings 3.5 - 5
- For the selected departments, the plot suggests that fewer students provide lower ratings, with those of 2 or less being relatively rare.



# Linear Models

Predicting overall rating using average GPA

| Ranking Features | Estimate | Standard Error | T-value | P-value |
|---|---|---|---|---|
| Overall Rating (Intercept) | 1.27771 | 0.16907 | 7.557 | 5.75e-14 |
| Average GPA | 0.72363 | 0.05193 | 13.935 | <2e-16 |

| Adjusted R-squared | P-value | Residual Standard Error |
|---|---|---|
| 0.07246 | 5.75e-14 | 0.9781 |

# Linear Models

Predicting easiness rating using average GPA

| Ranking Features | Estimate | Standard Error | T-value | P-value |
|---|---|---|---|---|
| Easiness (Intercept) | -0.69038 | 0.15837 | -4.359 | 1.36e-05 |
| Average GPA | 1.05632 | 0.04864 | 21.715 | <2e-16 |

| Adjusted R-squared | P-value | Residual Standard Error |
|---|---|---|
| 0.1599 | 1.36e-05 | 0.9163 |

# Linear Models

Non-stem departments BIC selection model using overall rating and easiness to predict GPA

| Ranking Features | Estimate | Standard Error | T-value | P-value |
|---|---|---|---|---|
| Average GPA (Intercept) | 2.85938 | 0.03999 | 71.505 | <2e-16 |
| Overall Rating | 0.04954 | 0.01298 | 3.815 | 0.000145 |
| Easiness | 0.12041 | 0.01233 | 9.768 | <2e-16 |

| Adjusted R-squared | P-value | Residual Standard Error |
|---|---|---|
| 0.2004 | < 2.2e-16 | 0.3169 |

# Linear Models

Stem departments BIC selection model using all rating categories to predict GPA

| Ranking Features | Estimate | Standard Error | T-value | P-value |
|---|---|---|---|---|
| Average GPA (Intercept) | 2.68210 | 0.03468 | 77.343 | <2e-16 |
| Overall Rating | 0.07535 | 0.01837 | 4.102 | 4.32e-05 |
| Easiness | 0.09649 | 0.01233 | 9.768 | <2e-16 |
| Clarity | –0.05302 | 0.01724 | -3.075 | 0.00214 |
| Workload | 0.04109 | 0.01364 | 3.011 | 0.00264 |

| Adjusted R-squared | P-value | Residual Standard Error |
|---|---|---|
| 0.1533 | < 2.2e-16 | 0.3366 |

# Linear Models

BIC model to predict GPA with Department Type as predictor

| Variable | Estimate | Standard Error | T-value | P-value |
|---|---|---|---|---|
| Average GPA (intercept) | 2.906221 | 0.027205 | 106.825 | < 2e-16 |
| Overall Rating | 0.083508 | 0.014670 | 5.692 | 1.4e-08 |
| Easiness | 0.125587 | 0.007773 | 16.157 | < 2e-16 |
| Clarity | -0.052294 | 0.013613 | -3.841 | 0.000125 |
| Dept.Typestem | -0.215825 | 0.013697 | -15.757 | < 2e-16 |

| Adjusted R-squared | P-value | Residual Standard Error |
|---|---|---|
| 0.2436 | < 2.2e-16 | 0.3294 |

# Sentiment Lexicon

**Phase 1** → Summarized texts longer than 512 tokens using Summarizer(), keeping English comments length longer than 1 only.

**Phase 2** → Identified positive, negative, and neutral sentiment using Roberta model.

**Phase 3** → Identified emotion sentiments (sadness, joy, love, anger, fear, surprise) using Bert model.

**Phase 4** → Conducted topic modeling for 3 topics (course lectures and materials, exams and homework, professor and study experience)

# Sentiment Data Analysis

- **Roberta sentiment distribution:**
  - Positive 0.569377
  - Negative 0.268722
  - Neutral 0.161901
- **Emotion Bert model sentiment distribution:**
  - joy 0.668511
  - anger 0.149863
  - sadness 0.094446
  - fear 0.036536
  - love 0.034171
  - surprise 0.016473
- **Topic modeling distribution:**
  - exams_homework 0.387244
  - prof_experience 0.336220
  - course_lec_material 0.276536

```
roberta_model
Negative        joy         0.337315
                anger       0.326470
                sadness     0.246149
                fear        0.064917
                love        0.012889
                surprise    0.012260
Neutral         joy         0.661884
                anger       0.208453
                sadness     0.055048
                fear        0.048265
                love        0.014088
                surprise    0.012262
Positive        joy         0.826706
                love        0.049926
                anger       0.049852
                sadness     0.034050
                fear        0.019807
                surprise    0.019659
Name: emotion_label, dtype: float64
```

# Sentiment Data Analysis

| Dept | Chem | Math | Physics | Stats | Comm | Econ | Poli Sci | Psych |
|---|---|---|---|---|---|---|---|---|
| Negative | 26.221434 | 26.906822 | 27.678896 | 29.803587 | 22.028207 | 34.699454 | 25.605701 | 23.295775 |
| Neutral | 19.562648 | 15.254578 | 14.892844 | 15.456874 | 14.842176 | 15.964091 | 13.919240 | 16.000000 |
| Positive | 54.215918 | 57.838599 | 57.428260 | 54.739539 | 63.129617 | 49.336456 | 60.475059 | 60.704225 |

P-value: ~0.0

| Topic | Lecture Material | Exams Home | Professor Experience |
|---|---|---|---|
| Negative | 25.523140 | 27.574171 | 27.173367 |
| Neutral | 15.381091 | 16.808464 | 16.143216 |
| Positive | 59.095769 | 55.617365 | 56.683417 |

P-value: 0.00063836325403836 84

| Department Type | non-STEM | STEM |
|---|---|---|
| Negative | 26.612405 | 27.052760 |
| Neutral | 15.361632 | 16.765695 |
| Positive | 58.025963 | 56.181545 |

P-value: 0.0044980375044490 72

# 03 Experiment

Comparing Machine Learning Models



# Machine Learning Models




## Decision Trees

A single predictive model that represents a flowchart-like structure, and each internal node represents a test on an attribute



## Random Forest

Structured by multiple decision trees during training and outputs the mode of individual tree predictions for classification tasks.



## KNN

A non-parametric classification algorithm that classifies new data points based on the majority of k nearest neighbors in the feature space.



## Log Regression

A statistical model for binary classification that estimates the probability of an event occurring based on chosen input features.

# Which variables are important in predicting Overall Ratings in STEM vs. Non-STEM classes?

## STEM

- Chemistry/ Biochemistry
- Statistics
- Mathematics
- Physics

## Non-STEM

- Psychology
- Economics
- Communications
- Political Science

## Interaction?

Overall Rating ~ (Easiness * Clarity) + (Easiness * Workload) + (Clarity * Workload) + Easiness + Clarity + Workload

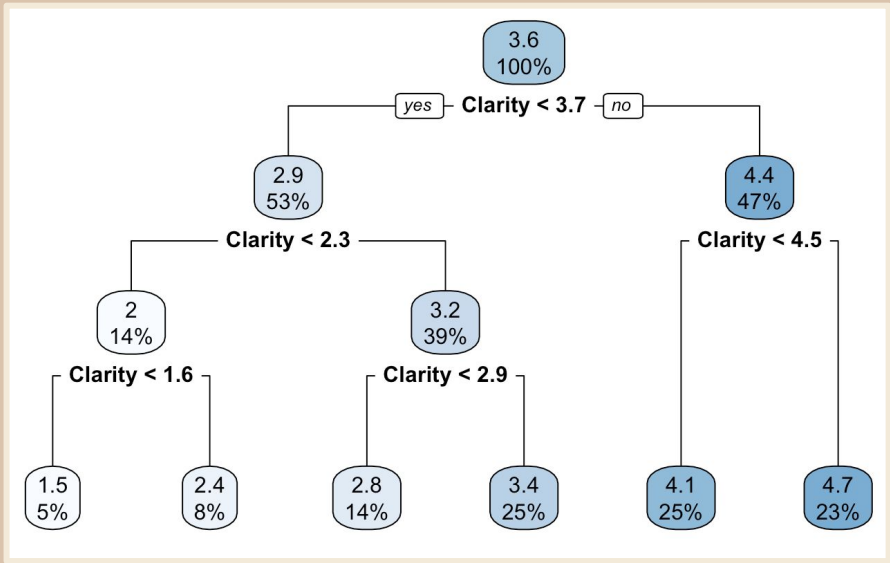
80% Training & 20% Testing

**Variable Importance**

| | |
|---|---|
| Easiness | 4.797140 |
| Clarity | 33.558240 |
| Workload | 2.051724 |
| Easiness:Clarity | 3.185241 |
| Easiness:Workload | 1.098494 |
| Clarity:Workload | 2.369941 |

# Decision Trees

## STEM



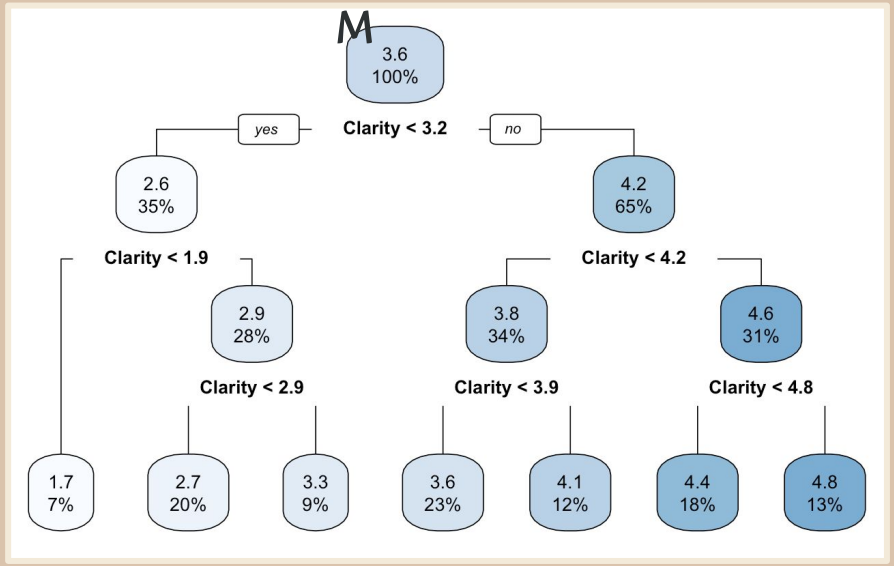

- There are 500 decision trees
- On average, the model's predictions are off by approximately 0.521 units
- R-Squared value of 0.761 suggests that approximately 76.1% of variability in Overall Rating can be explained by Clarity. It is a moderately fitting model.

RMSE: 0.521
R-squared: 0.761

# Decision Trees (ntrees = 500)

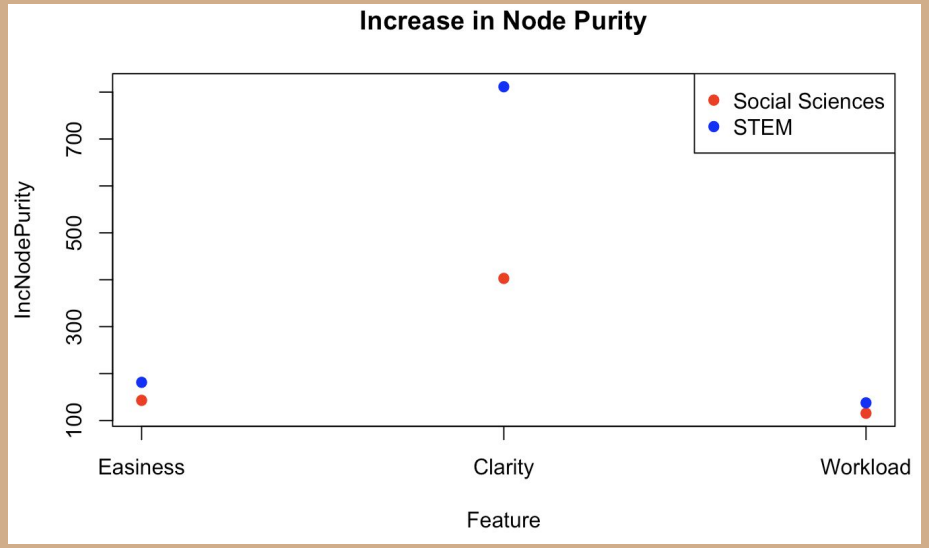
- There are 500 decision trees
- On average, the model's predictions are off by approximately 0.420 units
- R-Squared value of 0.822 suggests that approximately 82.2% of variability in the Overall Rating can be explained by Clarity. It is a moderately fitting model.

Non-STEM




RMSE: 0.420
R-squared: 0.822

# Random Forest (ntrees = 500)




- There are 500 decision trees
- Increase in Node Purity means that that specific feature is more effective in separating classes clearly
- For both Social Sciences and STEM courses, Easiness, Clarity, and Workload are the most important variables
- For STEM courses, Clarity plays a much larger role in predicting Overall Rating

**STEM**
RMSE: 0.553
R-squared: 0.733

**Non-STE**
**M**RMSE: 0.414
R-squared: 0.826

Based on the given Bruinwalk data, can we predict if a class is in a STEM or Non-STEM department?

# K-Nearest

| Actual/Predicted | Non-STEM | STEM |
|---|---|---|
| Non-STEM | 76 | 100 |
| STEM | 80 | 212 |

- Model predicts between two department types (non-STEM vs STEM)
- Omitted NA values within all department majors
- 70-30 split between training and testing data, respectively
- Chose k = 5 for classification of standardized values
- Model presents moderate bias towards specificity (true negative) given stronger sample bias towards STEM departments

# K-Nearest Neighbors

- Accuracy of 61.54% is relatively modest and can suggest that there is significant room for improvement.
- It also falls below the no information rate of 0.6667, which suggests that the model does not outperform a naive classifier.
- High P-value of 0.9913 suggests that there is no statistically significant evidence that the model's accuracy is better than random guessing based on majority class (STEM)

| Metric | Value |
|---|---|
| Accuracy | 0.6154 |
| 95% CI | (0.5696, 0.6597) |
| No Info Rate | 0.6667 |
| P-Value [ ACC > NIR] | 0.9913 |
| Kappa | 0.1615 |
| Mcnemar's Test P-Value | 0.1567 |
| Sensitivity | 0.4872 |
| Specificity | 0.6795 |
| Pos Pred Value | 0.4318 |
| Neg Pred Value | 0.7260 |
| Prevalence | 0.3333 |
| Detection Rate | 0.1624 |
| Detection Prevalence | 0.3761 |
| Balanced Accuracy | 0.5833 |
| Positive Class | non-stem |

# Logistic Regression

| Ranking Features | Estimate | Standard Error | T-value | P-value |
|---|---|---|---|---|
| Department Type (Intercept) | 5.7036 | 0.4792 | 11.902 | <2e-16 |
| Average GPA | -1.8000 | 0.1507 | -11.946 | <2e-16 |
| Clarity | -0.2134 | 0.1078 | -1.979 | 0.047848 |
| Overall Rating | 0.3854 | 0.1139 | 3.383 | 0.000717 |

| Null Deviance | P-value | Residual Deviance |
|---|---|---|
| 2301.1 | < 2.2e-16 | 2140 |

- Model predicts “Dept.Type” to distinguish between STEM and non-STEM classes.
- Reduced initial predictors from six to three (Avg_GPA, Clarity, Overall_Rating) after evaluating statistical relevance and multicollinearity.
- Non-STEM classes linked with higher average GPA and Overall Rating; STEM classes linked with higher Clarity levels.
- Attained ~68% accuracy, with better performance predicting the majority STEM category.

# Logistic Regression

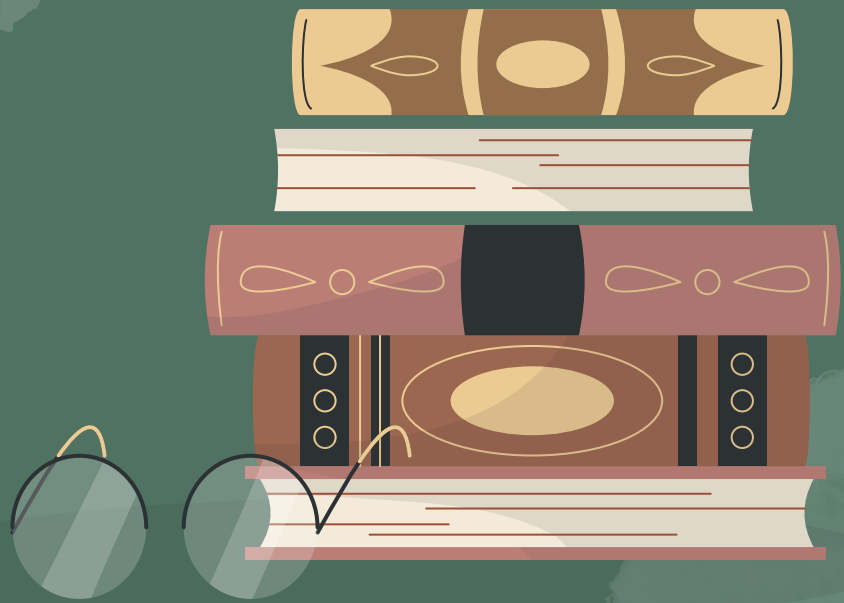
| Actual/Predicted | Non-STEM | STEM |
|---|---|---|
| Non-STEM | 112 | 71 |
| STEM | 169 | 389 |

- Regularization and attempts to fix class imbalance by training model on synthetically balanced dataset (SMOTE) did not yield improved accuracies.
- Future experiments might benefit by collecting more data from a diverse set of departments; this could possibly increase predictive power of classification models like logistic regression.

| Metric | Value |
|---|---|
| Accuracy | 0.6761 |
| Kappa | 0.2620 |
| Accuracy Lower | 0.6411 |
| Accuracy Upper | 0.7079 |
| Accuracy Null | 0.6208 |
| Accuracy P-Value | 0.00098 |
| McNemar P-Value | 0.00000038 |
| Sensitivity | 0.3986 |
| Specificity | 0.8457 |
| Positive Pred Value | 0.6120 |
| Negative Pred Value | 0.6971 |
| Precision | 0.6120 |
| Recall | 0.3986 |
| F1 | 0.4828 |
| Prevalence | 0.3792 |
| Detection Rate | 0.1511 |
| Detection Prevalence | 0.2470 |
| Balanced Accuracy | 0.6221 |

# 04 Conclusion

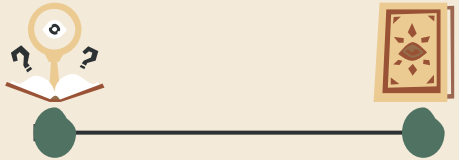
Comparing Machine Learning Models



# Research & Analysis

## *Important Variables*

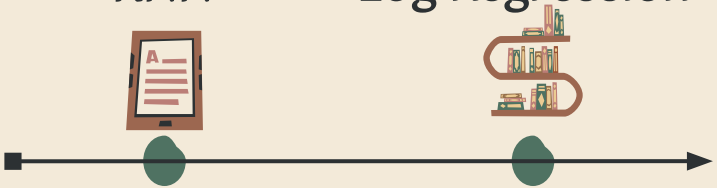
Decision Trees  Random Forest



- Clarity
- Easiness
- Workload

## *Predicting STEM vs. Non-STEM*

KNN  Log Regression



Accuracy: 62%  Accuracy: 68%

# Results & Analysis

- There were no significant differences in important variables in predicting overall ratings of courses
- Clarity was the most important factor in predicting a class' overall rating
- Non-STEM classes linked with higher average GPA and Overall Rating
- STEM classes linked with higher Clarity levels
- For predicting whether a class is in a STEM vs. Non-STEM Department, the Logistic regression model was preferred with a 68% accuracy





# Final Remarks

**Primary Objective**

- Utilize text mining, sentiment analysis, and machine learning to investigate significance of differences in predicting class overall ratings between North and South Campus
- Identify varying STEM against non-STEM departments
- Illuminate potential disparities in student satisfaction and instructional quality
- Enrich educational experiences for UCLA faculty and students

**Future Research/Recommendations**

- Explore longitudinal data to showcase evolving trends of influencing factors within educational experiences
- Utilize an unsupervised learning approach to predictive models
- Utilize arrays for more efficient computation regarding extract grade distributions
- Create an LLM or chatbot to generate class suggestions
- Create additional criteria such as lower/upper division courses



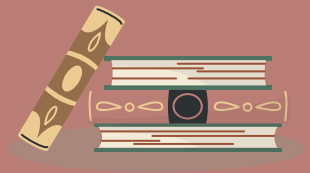




# Resources

- Bruinwalk
- UCLA Grade Distributions
- Python Request Library
- BeautifulSoup Documentation
- Sentiment Analysis: Concept, Analysis and Applications







# Thank you!

We would like to give our thanks to Professor Esfandiari for mentoring us throughout this project & Dr. Sugano and Dr. Zhang for giving us this opportunity to further ourselves as consultants and statisticians.

